DECK 13 · WEEK 4 · MORNING 1

# Fixed Income I
# Instruments, Markets & Issuance

*How issuers finance themselves and investors get paid — from indentures to auctions*

CFA Level I · Vol 6 Learning Modules 1–5

# Roadmap — Fixed Income I — Instruments, Markets & Issuance

*5 sections · ~2 hours · Built bottom-up from CFA readings*

**01** **Bond features & indenture**
Six bond features · fixed / floating / zero coupons · FRN math · affirmative vs negative covenants · callable bonds (VIVU schedule)

**02** **Cash-flow structures**
Bullet vs amortizing vs balloon · USD 400k 30y mortgage · sinking funds · waterfalls · FRN / step-up / PIK / TIPS · puts & convertibles (ZTGB)

**03** **Issuance & trading mechanics**
Credit × maturity segmentation · S&P ratings · bond indexes · IG corporate issuance timeline · OTC secondary markets

**04** **Corporate FI markets**
Lines of credit · commercial paper & ABCP · repo mechanics + haircut · reverse-repo short-selling · Bear Stearns · IG vs HY

**05** **Government & supranational issuers**
Economic balance sheet · Ricardian equivalence · single- vs multiple-price auctions · agencies · supranationals

# Learning objectives

*By the end of this deck you should be able to:*

1. Describe the six features of a fixed-income security (issuer, maturity, principal, coupon, seniority, contingency provisions) and read them off a prospectus summary.

2. Distinguish bullet, amortizing, and balloon cash-flow structures; compute the level payment of a fully amortizing loan and split it into interest and principal.

3. Contrast affirmative and negative covenants; identify the incremental covenant stack imposed on high-yield versus investment-grade issuers.

4. Compute the purchase price, haircut, and repurchase price of a repo given an initial margin, and say who keeps ownership of the collateral.

5. Compare single-price and multiple-price sovereign auctions; explain primary-dealer obligations and the "non-economic" demand that compresses sovereign yields.

SECTION 01

# Bond features & indenture

# Every bond is fully specified by six features — issuer, maturity, principal, coupon, seniority, and contingency provisions — spelled out in the prospectus

*Worked reference — BRWA Corporation 3.2% Five-Year Notes, USD 300m par, senior unsecured, semi-annual coupon → USD 4.8m every six months.*

| Feature | What it defines | BRWA 3.2% Five-Year Notes |
|---|---|---|
| Issuer | Legal entity liable for interest and principal | Bright Wheels Automotive |
| Maturity | Date of final payment; tenor = remaining life | 5 years from settlement |
| Principal (par) | Amount repaid at maturity | USD 300,000,000 |
| Coupon rate & frequency | Interest schedule — fixed, floating, or zero | 3.2% fixed, semi-annual |
| Seniority | Ranking vs other issuer debt in liquidation | Senior unsecured |
| Contingency provisions | Embedded options: call, put, or conversion | None (plain vanilla) |

**ANNUAL COUPON = PAR × COUPON RATE**

USD 300m × 3.2% = USD 9.6m per year, paid semi-annually as USD 4.8m. At maturity the investor receives the final USD 4.8m coupon PLUS the USD 300m principal, for a total of USD 304.8m — the one cash flow that is not level.

[1] Current yield = annual coupon / price. At USD 101 per 100 of face, BRWA's current yield is 3.2% / 1.01 = 3.168%.
[2] YTM is the IRR of price against promised cash flows: at 101, r = 1.49% semi-annually, or 2.98% annualised.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1, "Features of Fixed-Income Securities" and "Yield Measures", pp. 7–13.*

# Coupons come in three flavours — fixed, floating, or zero — and a floater's rate is always MRR plus a spread fixed at issuance

*The credit spread is set at issuance and never moves; only the MRR resets. So an FRN tracks short-term rates while keeping a fixed credit cushion.*

$$\text{FRN coupon} = \text{MRR} + \text{Credit spread} \qquad \text{Quarterly interest} = (\text{Coupon} \times \text{Principal}) / 4$$

*where: MRR = market reference rate (SOFR, €STR, …), reset each period. The credit spread is issuer-specific and fixed for the life of the note.*

## CFA LM 1 — ANTELAS AG FRN · EUR 250m · MRR + 250 bps · QUARTERLY

**1.** MRR for the final interest period = 0.25% → coupon rate = 0.25% + 2.50% = 2.75% p.a.
**2.** Annual interest = EUR 250,000,000 × 2.75% = EUR 6,875,000
**3.** Quarterly interest = 6,875,000 / 4 = EUR 1,718,750
**4.** Maturity-date payment = principal 250,000,000 + final coupon 1,718,750 = EUR 251,718,750
**5.** If MRR resets to 0.50%, coupon goes to 3.00% → EUR 1,875,000. The 250 bps spread does not move.

**Quarterly coupon EUR 1,718,750 · final payment EUR 251,718,750 · spread fixed at 250 bps**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1 Practice Problems (Antelas AG FRN), pp. 20–22; Learning Module 2, Exhibit 6, p. 33.*

# The indenture is the legal contract; covenants split into affirmative (must-do) and negative (must-not-do), and tighten sharply as credit quality falls

*BRWA is investment grade and gets a short list. VIVU is high-yield secured and wears a full monitoring regime — the covenant stack is the credit premium made legal.*

## B R W A · I G · U N S E C U R E D

AFFIRMATIVE — must do:

- Use of proceeds defined
- Pari passu with unsecured debt
- Timely financial reporting
- Cross-default clause

NEGATIVE — must not do (light):

- Limitation on liens
- Limitation on sale & leaseback
- Merger / consolidation limits

No leverage test. No dividend block. No call.

## V I V U · H Y · S E C U R E D C A L L A B L E

All of the above, PLUS a security pledge:

- Restriction on additional debt
- Dividend / distribution block
- Net debt / EBITDA ≤ 5.00×
- Interest coverage ≥ 2.50×

INCURRENCE TEST — tighter, at 4.50× and 3.00× — must be met before paying a dividend or issuing new debt.

Plus a fixed-price call schedule: the issuer keeps the refinancing option.

A breach gives bondholders recourse: a change in financial terms (higher rate or added security), accelerated payment, or termination of the agreement.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 1, "Bond Indentures and Covenants", Exhibits 8–10, pp. 14–19.*

# A call belongs to the ISSUER — it buys back at a fixed schedule when rates fall, which caps the investor's price and pays the investor a higher yield up front

*The call price declines toward par as maturity approaches: the option is worth less with less time left, so VIVU pays less to exercise it.*

| Window | Call price (% par) |
|---|---|
| Years 0–3 (protection) | Non-callable |
| Years 3–4 | 103.25% |
| Years 4–5 | 102.50% |
| Years 5–6 | 101.75% |
| Years 6–7 | 101.00% |
| At maturity | 100.00% |

## WHY THE CALLABLE YIELDS MORE

If YTM rises ABOVE the coupon, the call is worthless and the bond behaves like a straight bond. If YTM falls BELOW the coupon, the price is capped at the call price while a non-callable keeps rising. Investors demand a higher yield up front for that capped upside — call risk.

## ILLUSTRATION — NOT THE CURRICULUM

When does VIVU call? 6.5% seven-year, USD 400m; yields fall 200 bps by year 3, so VIVU can refinance at 4.5%.

**1.** Call protection ends at year 3 → call price 103.25% = USD 413m
**2.** Call premium paid = 3.25% × 400m = USD 13m
**3.** Annual interest saved = 400 × (6.5% − 4.5%) = USD 8m
**4.** Premium recovered in 13 / 8 = 1.6 years, well inside the remaining four → VIVU calls
**5.** Investor is repaid early and reinvests at 4.5%: reinvestment risk, realised.

**Issuer refinances cheaply; investor gets capped upside and reinvestment risk**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, "Fixed-Income Contingency Provisions", Exhibit 9 (VIVU), pp. 39–40.*

SECTION 02

# Cash-flow structures

# Bullet, amortizing, and balloon bonds differ only in WHEN principal comes back — same issuer, same rate, three different cash-flow shapes

*Same five-year term, USD 300m par, 3.2% semi-annual rate. Only the principal schedule changes — and it changes everything the investor holds.*

| Period (years) | Bullet bond | Partial + 150m balloon | Fully amortizing |
|---|---|---|---|
| 0 (issue) | −300.0 | −300.0 | −300.0 |
| 0.5 | +4.8 | +18.75 | +32.70 |
| 1.0 – 4.5 each | +4.8 | +18.75 | +32.70 |
| 5.0 (maturity) | +304.8 | +168.75 | +32.70 |
| **Principal** | **At maturity** | **Partly, plus balloon** | **Over the whole life** |

### BULLET

Interest-only coupons, principal in one lump. The standard corporate and government bond. Highest reinvestment risk at maturity.

### FULLY AMORTIZING

$A = 0.016 \times 300 / [1 - 1.016^{-10}]$ = USD 32.70 each period. Credit risk falls as the balance shrinks; reinvestment risk rises.

### PARTIAL + BALLOON

PV of the 150 balloon = 127.98, so 172 amortizes: $A = 0.016 \times 172 / [1 - 1.016^{-10}] = 18.75$. Final payment 18.75 + 150.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, "Fixed-Income Cash Flow Structures", Exhibits 1–4, pp. 27–31.*

# A fully-amortizing payment is an annuity solved for PMT — early payments are mostly interest, late payments are almost entirely principal

*Same identity as the PV of an annuity, rearranged: given the loan, the periodic rate, and the number of periods, solve for the level payment.*

$$A = r \times Principal / [1 - (1 + r)^{(-N)}]$$

*where: A = level payment; r = periodic rate; N = number of payments. Interest each period = r × outstanding balance.*

## CFA LM 2 EXAMPLE 1 — USD 400,000 MORTGAGE

**1.** Loan 400,000 on a 500,000 home; r = 3.50% / 12 = 0.29167%; N = 360 months
**2.** Numerator: r × P = 0.0029167 × 400,000 = 1,166.67
**3.** Denominator: $1 - (1.0029167)^{-360} = 1 - 0.35047 = 0.64953$
**4.** A = 1,166.67 / 0.64953 = USD 1,796.18 per month
**5.** Month 1 = 1,166.67 interest + 629.51 principal — 65% of it is interest
**6.** Total paid = 1,796.18 × 360 = USD 646,625

**USD 1,796.18 / month · USD 246,625 of interest on USD 400,000 borrowed**

| Month | Interest | Principal | Balance |
|---|---|---|---|
| 1 | 1,166.67 | 629.51 | 399,370 |
| 2 | 1,164.83 | 631.35 | 398,739 |
| 3 | 1,162.99 | 633.19 | 398,106 |
| … | ... | ... | ... |
| 358 | 15.63 | 1,780.55 | 3,577 |
| 359 | 10.43 | 1,785.75 | 1,791 |
| 360 | 5.22 | 1,790.96 | 0 |

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, Example 1 — 30-Year Fixed-Rate Residential Mortgage Loan, p. 31.*

# Sinking funds retire principal through the life of the bond; ABS waterfalls repay principal sequentially A → B → C while interest goes to everyone

*Both cut credit risk for the senior investor and raise reinvestment risk — but by completely different mechanisms: one shrinks the balance, one ranks the claims.*

## SINKING FUND

The issuer sets funds aside over time in an escrow account to retire a predetermined amount of principal early, on terms agreed at issuance.

How the bonds are taken out:

- The trustee redeems a set amount, selected from investors at random
- Or the issuer repurchases at a predetermined fixed price

Effect: principal outstanding falls before maturity, so credit risk falls and reinvestment risk rises — exactly like amortization.

## WATERFALL (ABS / MBS)

INTEREST is paid to all classes with no preference. Only PRINCIPAL is sequential:

1. Tranche A receives principal first
2. Tranche B only once A is fully repaid
3. Tranche C only once A and B are repaid

Shortfalls are borne bottom-up, so Tranche A faces the lowest credit risk, then B, then C. Standard in RMBS, CMBS and CLO structures.

Tranche A coupon < B < C. The extra coupon is payment for standing further back in the principal queue.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, sinking funds and waterfall structures, Exhibit 5, p. 32.*

# Floating, step-up, credit-linked, payment-in-kind and index-linked coupons are five ways to make the interest stream adapt instead of staying fixed

*Each mechanism moves one specific risk — interest rate, ESG performance, credit deterioration, cash strain or inflation — between issuer and investor.*

| Structure | Coupon mechanism | Typical use |
|---|---|---|
| Floating-rate note (FRN) | MRR + fixed spread, reset each period | Bank loan assets |
| Step-up bond | Rises by preset margins on preset dates or on a trigger | ESG-linked bonds |
| Credit-linked coupon | Spread steps with leverage: Antelas 250→325 bps | Leveraged loans |
| Payment-in-kind (PIK) | Interest paid as new principal — no cash out | Levered LBOs |
| Index-linked (TIPS) | Principal indexed to CPI; coupon on adj. principal | Inflation hedge |

### CFA LM 2 EXAMPLE 2 — TIPS CASH FLOW

USD 115m principal · 1.00% p.a. coupon paid semi-annually · CPI rises 265.00 → 267.25, i.e. +0.849%.

**Adjusted principal 115,976,350 → annual 1,159,764 → semi-annual coupon USD 579,882**

[1] Capital-indexed (TIPS): principal is CPI-adjusted; at maturity investors usually receive the greater of adjusted principal or par.
[2] Interest-indexed bonds keep nominal principal fixed and index only the coupon — an FRN whose reference rate is inflation.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, variable interest and index-linked structures (Antelas leveraged loan, PPC SLB, TIPS Example 2), pp. 32–36.*

# Zero-coupon bonds replace every coupon with a single discount at issuance; deferred-coupon bonds just push the coupons back in time

*Both pay nothing early on — but their motives, their pricing and the risks they carry are different animals entirely.*

## ZERO-COUPON

One cash flow: par at maturity. Priced below par whenever rates are positive — the discount IS the cumulative interest.

Why investors want them:

- No coupon to reinvest, so no reinvestment risk
- Clean matching of a known future obligation
- Natural fit for pension and insurance ALM

Where they come from:

- Governments at tenors under 12 months (T-bills)
- Intermediaries stripping coupon and principal payments off sovereign bullets
- Longer maturities became common after 2008 as sovereign rates fell to or below zero

## DEFERRED-COUPON

No interest for the first few years, then a higher coupon through to maturity. A zero-coupon bond is the extreme case.

Why issuers use them:

- Conserve cash immediately after issuance
- Common where a project generates no income until it is finished — construction, infrastructure
- Often signals weaker credit quality

Pricing:

- Typically at a discount to par
- The later, higher coupons do not usually make up for the interest forgone in the early years

Both are still priced as the present value of known future cash flows — the same bond maths as Deck 01. Nothing new, only a different schedule.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, zero-coupon and deferred-coupon structures, Exhibit 8, pp. 36–37.*

# Put and conversion provisions belong to the INVESTOR — a put floors the price, conversion swaps debt for equity, and both cut the yield the issuer pays

*Options held by the investor are paid for in yield: a putable bond yields LESS than an option-free bond, and a convertible yields less still.*

## CFA LM 2 EXHIBIT 10 — ZTG BIOTECH 1.25% FIVE-YEAR CONVERTIBLE

EUR 300m issue, unsecured, traded in EUR 1,000 units. Share price at issue EUR 28.00; conversion price EUR 42.00.

**1.** Conversion ratio = par / conversion price = 1,000 / 42 = 23.81 shares per unit
**2.** Conversion premium = (42 − 28) / 28 = 50.0% over the share price at issue
**3.** A year later the shares trade at EUR 35 → conversion value = 23.81 × 35 = EUR 833.35
**4.** The bond trades near par, so the investor holds, takes the 1.25%, and waits for the shares to clear 42.

**Convert only once the conversion value beats the bond – i.e. share price above EUR 42**

### PUTABLE BOND

If rates rise and the price falls, the investor sells the bond back at the put price — usually par. That price is a floor, and the issuer is compensated with a lower coupon.

### CONTINGENT CONVERTIBLE (CoCo)

Converts on the DOWNSIDE, automatically, if a bank's core capital ratio breaches its regulatory minimum. Not the investor's choice — hence a higher yield.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 2, putable, convertible and contingent convertible bonds, Exhibit 10 (ZTG BioTech), pp. 40–42.*

SECTION 03

# Issuance & trading

# Fixed-income markets are sliced three ways at once — issuer type, credit quality, and time to maturity — and every instrument lives in a specific cell

*Each cell has a natural issuer and a natural investor. Mismatches between the two — short-term funding against long-term assets — are where blow-ups come from.*

| Credit quality | Short-term (<1y) | Intermediate (1-10y) | Long-term (>10y) |
|---|---|---|---|
| Default risk-free (DM sov) | Treasury bills | Treasury notes | Treasury bonds |
| Investment grade corp | Repo · CP · ABCP | Unsecured bonds · ABS | Unsecured bonds · MBS |
| High yield corp | — (no access) | Secured bonds · lev loans | Secured corporate bonds |

## THE NATURAL INVESTOR IN EACH CELL

Money-market funds and corporate treasuries → short-dated risk-free and IG. Financial intermediaries and central banks → across the risk-free curve. Bond funds and ETFs → intermediate IG. Pension funds and insurers → long-dated risk-free and IG, matching liabilities. Asset managers → long IG. Hedge funds and distressed funds → high yield.

## ONE ISSUER, MANY INSTRUMENTS

An equity issuer typically has one instrument outstanding. At the end of 2021 Apple had over 80 fixed-income instruments and a single common stock.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Fixed-Income Segments, Issuers, and Investors", Exhibits 1–2, pp. 57–61.*

# Credit ratings draw a bright line at BBB-/Baa3 — at or above it, investment grade; at BB+/Ba1 or below, high yield, speculative grade, or junk

*That line is regulatory, contractual and psychological all at once — many institutional mandates are hard-capped at investment grade, so crossing it forces selling.*

| Band | S&P | What S&P means by it | Who lives there |
|---|---|---|---|
| IG — highest | AAA | Extremely strong capacity to pay | DM sovereigns, SSAs |
| IG — upper | AA / A | Very strong / strong; some sensitivity | Large IG corporates |
| IG — lower | BBB+ to BBB- | Adequate; more exposed to downturns | Core of IG indices |
| HY — upper | BB+ / BB / B | Faces major ongoing uncertainties | Leveraged finance |
| HY — lower | CCC / CC / C | Vulnerable; needs favourable conditions | Distressed funds |
| Default | D | Payment default or bankruptcy filing | Restructuring |

### FALLEN ANGEL

A former investment-grade issuer downgraded to high yield. Its bonds still carry IG documentation — long maturity, few covenants — but mandate-constrained holders become forced sellers.

### WHAT A RATING MEASURES

Both the probability of default AND the expected loss if default occurs. Agencies rate issuers and individual issues separately, and many mandates cap unrated holdings.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, S&P credit ratings box and IG / HY definitions, pp. 59–60.*

# Bond indexes differ from equity indexes in three ways — far more constituents, far higher turnover, and weighting by market value of debt outstanding

*One issuer, many eligible bonds — some indexes carry over 10,000 constituents. Bonds mature and get refinanced, so indexes rebalance monthly and funds hold a representative sample, never the full basket.*

## GLOBAL AGGREGATE

Scope: fixed-rate bonds from sovereign, government, corporate and securitized issuers across 28 developed and emerging markets.

Criteria:

- Investment-grade rating
- At least 1y to maturity
- Minimum issue size by market
- Fully taxable issues only

Excludes: high yield, unrated, and anything below minimum size.

Rebalanced monthly.

## J.P. MORGAN EMBI+

Scope: USD-denominated debt of emerging market sovereign issuers.

Criteria:

- Sovereign rated Baa1 / BBB+ or below
- USD only; no FX-linked payments
- USD 500m minimum outstanding
- At least 2.5y to maturity

Purpose: a higher dollar return than DM sovereigns, in exchange for the external debt risk of the issuer country.

Rebalanced monthly.

## MSCI EUR CORP SRI

Scope: euro-denominated IG corporate bonds with an ESG overlay.

Criteria:

- Rated Baa3 / BBB- or above
- MSCI ESG rating of BBB or higher
- EUR 300m minimum outstanding
- At least 1y to maturity

Excludes: alcohol, tobacco, gambling, adult entertainment, GMOs, nuclear, firearms, weapons, thermal coal, oil sands — and any issuer with a "Red" MSCI controversy score.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Fixed-Income Indexes", Exhibits 3–5, pp. 65–68.*

# Investment-grade corporate issuance is compressed into a single trading day — announced at 9:15, priced at 3pm, free to trade the same or next day

*Frequent issuers work off a shelf registration and pick the day opportunistically. The roadshow is an electronic file released at announcement plus one investor call.*

| Time | Event | Book status |
|---|---|---|
| 9:00 am | Underwriters and issuer agree to launch | Pre-launch |
| 9:15 am | Announced, size / spread / tenor TBD; e-roadshow | Book opens |
| 10:30 am | Investor conference call with issuer + underwriters | Book building |
| 12:00 noon | Price guidance out; order book CLOSED to investors | Book closed |
| 1:00 pm | Transaction LAUNCHED — final size, price, tenors | Launched |
| 1:00–3:00 pm | Underwriters allocate the transaction to investors | Allocation |
| 3:00 pm | Priced — government benchmark and coupon set | Priced |
| 4:00 pm | Final term sheet; bonds free to trade same/next day | Secondary |

**THE CONTRAST — HIGH YIELD, SECURED, AND PRIVATE**

HY and secured deals run far longer: complex covenants, collateral diligence, and often a best-efforts sale rather than an underwritten one. A private placement skips the public process entirely, selling to a small group of investors.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Primary and Secondary Fixed-Income Markets", Exhibit 6, pp. 70–71.*

# Secondary bond markets are quote-driven and over-the-counter; the bid-offer spread is the honest measure of how liquid a line really is

*Unlike equities, bonds rarely trade on an exchange — price discovery lives at dealer desks, and liquidity varies even between two bonds of the same issuer.*

| Bond type | Bid-offer spread | Why |
|---|---|---|
| On-the-run DM sovereign (latest UST 10y) | Fraction of 1 bp | Primary dealers quote in size |
| Recent bond, frequent IG issuer | A few bp | Dealers hold trading inventory |
| Seasoned or infrequent IG corporate | 10–20 bp or more | Rarely traded; small sizes only |
| Distressed debt (issuer near bankruptcy) | Wide; price << par | Forced sellers meet opportunists |
| Bonds that do not trade at all | No live quote | Priced off comparable liquid bonds |

## HERTZ 2020 — WHAT DISTRESS LOOKS LIKE IN THE TAPE

Hertz filed for Chapter 11 in May 2020 as COVID hit car rental. Its 2022 and 2028 bonds traded below 10% of par, down more than 90% in three months — then recovered to pre-pandemic prices a year later, when a reorganisation with outside investors left them intact. The curriculum's general point: distressed bonds keep trading until the issuer liquidates or restructures, whereas by that stage the equity has usually already been delisted.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 3, "Secondary Fixed-Income Markets" and Example 4 (Hertz), pp. 72–73.*

SECTION 04

# Corporate FI markets

# Short-term corporate funding runs from cheap-and-uncommitted bank lines up to binding multi-year revolvers with covenants — reliability is bought with cost

*Climb the reliability ladder and the bank starts asking for an upfront fee and covenants in exchange for liquidity it is legally bound to provide.*

| Facility | Reliability | Bank commitment | Cost to borrower | Covenants |
|---|---|---|---|---|
| Uncommitted line of credit | Lowest | Bank may decline | MRR + spread, drawn | None |
| Committed line (364-day) | Medium | Written commitment | + fee, e.g. 0.50% | Rare |
| Revolver (multi-year) | Highest | Binding, multi-year | + fee + covenants | Full set |
| Secured / asset-based loan | Credit-dependent | Collateral pledged | Rate reflects lien | Per collateral |
| Factoring of receivables | Not a facility | Receivables sold | Discount to face | None |

## THE RULE OF THUMB

A strong IG borrower with stable deposits at the bank takes an uncommitted line — no fee. A weaker borrower needs a revolver: binding liquidity, paid for with a fee and covenants. Revolvers also backstop commercial-paper programmes.

## ASSIGNMENT vs FACTORING

Assignment pledges receivables as collateral and the company still does the collecting. Factoring SELLS them at a discount, and the factor takes over credit granting and collection.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Short-Term Funding Alternatives", Exhibit 1, pp. 80–82.*

# Commercial paper is unsecured, typically under three months, and rolled over constantly — the rollover risk is plugged by a committed backup liquidity line

*Roughly 60% of annual CP issuance comes from financial institutions. ABCP wraps short-term loans into an SPE and keeps them off the sponsoring bank's balance sheet.*

## CP — UNSECURED

Issued by large, highly rated corporates and financial institutions, publicly or by private placement.

- Usually under three months
- An unsecured promissory note
- Repaid at maturity with newly issued CP — "rolled over"
- Rollover risk plugged by a committed backup liquidity line from banks
- Eurocommercial paper is the international cousin: smaller tickets, less liquid than USCP
- Short maturity means the market reprices fast; defaults are rare

## ABCP — SECURED

A bank transfers short-term loans or receivables to a special purpose entity.

- The SPE — not the bank — issues the CP, backed by those assets
- The bank provides a backup liquidity line, not a guarantee
- Off balance sheet for the bank, so lower capital cost
- Investors buy exposure to a loan pool they could not otherwise reach
- Grew fast pre-2008; when issuers could not roll, numerous SPEs failed

## CDs & INTERBANK

Certificates of deposit:

- Non-negotiable — early withdrawal carries a penalty
- Negotiable — sold in the market before maturity
- Large-denomination CDs are core wholesale funding

Interbank market: secured or unsecured, overnight out to a year, priced off the MRR.

Central bank funds market moves reserves between banks at the central bank funds rate. The discount window is the last resort — collateralised, dearer, and it invites supervision.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, external security-based financing, bank funding, ABCP (Exhibit 3), pp. 82–85.*

# A repo is a collateralised loan dressed as a sale — the repo rate is the interest, the haircut is the overcollateralisation, and the SELLER keeps ownership

*Legally a sale plus a forward repurchase; economically a secured loan. Initial margin above 100% means the cash lender is overcollateralised — the haircut is the borrower's own money.*

$$\text{Initial margin} = \text{Sec price} / \text{Purchase price} \qquad \text{Haircut} = (\text{Sec price} - \text{Purchase price}) / \text{Sec price}$$

*where: Repurchase price = Purchase price × [1 + r × d/360]. Variation margin = (Initial margin × Purchase price$_t$) − Security price$_t$*

## CFA LM 4 KNOWLEDGE CHECK — 30-DAY REPO ON USD 100m UST

**1.** Terms: USD 100m 5y UST, 30-day repo, 0.25% repo rate, 102% initial margin.
**2.** Seller (cash borrower) delivers the Treasuries; buyer (cash lender) sends the cash.
**3.** Purchase price = Security price / Initial margin = 100,000,000 / 1.02 = USD 98,039,216
**4.** Haircut = (100,000,000 − 98,039,216) / 100,000,000 = 1.96%; the seller funds 1,960,784 elsewhere
**5.** Repurchase price = 98,039,216 × [1 + (0.25% × 30 / 360)] = USD 98,059,641
**6.** Repo interest to the cash lender = 98,059,641 − 98,039,216 = USD 20,425

**Lends 98.04m against 100m of UST for 30 days · haircut 1.96% · repo interest 20,425**

[1] The cash borrower (security seller) keeps ownership of the collateral and its coupons; the cash lender receives only the repo rate.
[2] General collateral repo references a pool of eligible bonds; a special trade names one security and prices off the special rate.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Repurchase Agreements", Exhibit 4 and Knowledge Checks, pp. 86–89.*

# Seen from the cash lender's side the same trade is a reverse repo — and it is how a hedge fund borrows a bond in order to short it

*Here the hedge fund is the security BUYER and cash LENDER — so it earns the repo rate. Its motive is not the interest; it is renting the specific bond.*

| Step | Time | Hedge-fund action | Cash flow |
|---|---|---|---|
| 1 | t = 0 | Borrow USD 100m 5y UST in repo; deliver 100m of cash | −100m cash, +UST |
| 2 | t = 0 | Simultaneously sell the UST in the cash market at par | +100m cash, −UST |
| 3 | t = T | Buy back the UST in the cash market at 99% of face | −99m cash, +UST |
| 4 | t = T | Deliver the UST back; receive repurchase price 100.02m | +100.02m cash, −UST |
| **Net** | **30 days** | **Price gain 1.00m + repo interest 0.02m received** | **USD 1,020,833** |

## SPECIAL COLLATERAL

A trade that names one specific security is a "special". Heavy short demand pushes its repo rate below the general-collateral rate — sometimes below zero, in which case the cash lender pays interest to the cash borrower.

## BILATERAL vs TRIPARTY

Bilateral repo settles directly between the two parties. Under triparty, a custodian holds cash and collateral and handles valuation — cheaper and safer operationally, but the credit risk between the parties is unchanged.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, Exhibit 5 and the reverse repo Knowledge Check, pp. 90–91.*

# Heavy reliance on repo funding killed Bear Stearns — USD 18bn of cash became USD 2bn in two days when counterparties simply declined to roll

*The lesson is structural, not moral: repo is cheap because it is uncommitted, and being collateralised does not stop the funding from vanishing.*

| Date | What happened | Signal |
|---|---|---|
| End-2006 | USD 350bn assets, USD 9bn revenue — 5th-largest US IB | Healthy |
| July 2007 | MBS market deteriorates; sponsored subprime funds fail | Amber |
| End-2007 | First ever quarterly loss; >USD 2bn writedown; downgrade | Amber |
| 2006 → 2007 | Unsecured ST funding 25.8→11.6bn; CP 20.7→3.9bn | Amber |
| 10 March 2008 | Cash position of USD 18bn | Red |
| 12 March 2008 | Cash USD 2bn — repo lenders will not roll; PB clients flee | Red |
| 14 March 2008 | Emergency Fed loan against securities collateral | Rescue |
| 17 March 2008 | Agrees to be bought by JPMorgan, which takes a Fed loan | Failed |

[1] Repo is uncommitted and mostly overnight: the cash lender need not roll. Collateral does not save a firm that has lost market confidence.

[2] Repo risks the curriculum names: default, collateral, margining, legal, and netting / settlement — plus rollover and liquidity risk.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, repo risks, triparty repo, and "The Fall of Bear Stearns and the Repo Market", pp. 92–94.*

# Under normal conditions a longer maturity means a higher benchmark yield AND a wider credit spread for the same issuer — the credit premium widens with time

*Issuer and investor sit on opposite sides of the same trade-off, and both errors have names: price risk and reinvestment risk for one, rollover risk for the other.*

| Maturity | Government YTM | BRWA YTM | BRWA credit spread |
|---|---|---|---|
| 3-year | Below the 5y point | Below the 5y point | Narrower than 90 bps |
| 5-year | 2.30% | 3.20% | 90 bps |
| 7-year | Above the 5y point | Above the 5y point | Wider than 90 bps |

### INVESTOR WITH A FIVE-YEAR HORIZON

Buys the 7-year for its higher YTM → price risk: it must sell what will then be a two-year bond, at an unknown price.

Buys the 3-year instead → reinvestment risk: coupons and principal come back early and must be reinvested at unknown rates.

### ISSUER WITH A FIVE-YEAR PROJECT

Issues 3-year for the lower YTM → rollover risk: BRWA must refinance in three years at whatever the market then charges.

Issues 7-year → pays the term premium and the wider spread for two years longer than it needs the money.

[1] Only the five-year point is numeric in Exhibit 7. The 3y and 7y entries are directional — curve and spread both rise with maturity.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Long-Term Corporate Debt", Exhibit 7 (BRWA maturity choices), pp. 95–96.*

# Investment grade rides the government benchmark with a thin spread and light covenants; high yield has equity-like cash flows and a full covenant leash

*The YTM gap is not just "more default risk" — it marks a completely different contractual, analytical and behavioural regime.*

| Dimension | Investment grade (BRWA) | High yield (VIVU) |
|---|---|---|
| YTM composition | Mostly the benchmark; thin spread | Wide issuer-specific spread |
| Cash flow profile | Bond-like — payments highly likely | Equity-like — cash flows uncertain |
| Analytical approach | Ratios and rating migration | PD, loss given default, collateral |
| Covenant stack | Light — liens, leaseback, M&A | Heavy — leverage, coverage, payouts |
| Repayment sources | Operating cash flow only | Cash flow plus pledged assets |
| Maturity range | Up to 30 years | Typically 10 years or less |
| Contingency features | Make-whole call, or none | Fixed-price call schedule |
| Market access | Always open; opportunistic refi | Cyclical; restructure to refinance |

**WHY HIGH-YIELD ISSUERS PAY FOR THE CALL**

An issuer that expects its own credit to improve buys the call: the option gains value exactly as its borrowing cost falls. IG issuers stagger maturities instead — the same problem, solved with a ladder rather than an option.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 4, "Long-Term Corporate Debt", Exhibits 8–10 (BRWA vs VIVU), pp. 96–99.*

SECTION 05

# Government issuers

# A sovereign's economic balance sheet is the PV of expected tax revenues against promised expenditures — fiscal policy sets the level of debt, debt management sets its composition

*Public accounting is usually cash-based: it leaves out depreciation on public assets and unfunded pension accruals. The economic balance sheet puts expected future claims and obligations back in.*

## THE ECONOMIC BALANCE SHEET

CLAIMS

- Short-term financial claims and assets
- Fixed assets — infrastructure, land, SOEs
- FX, commodity and other reserves
- PV of expected future tax revenues

OBLIGATIONS

- Domestic currency outstanding
- Short- and long-term debt, domestic and FX
- Pension and other domestic obligations
- Supranational and external obligations
- PV of promised future expenditures

## LEVEL vs COMPOSITION

FISCAL POLICY sets the LEVEL of debt.

- Deficits raise borrowing; surpluses cut it
- Sensitive to growth and inflation forecasts

DEBT MANAGEMENT sets the COMPOSITION.

- Short vs long; fixed, floating or linker
- Domestic vs foreign currency

RICARDIAN EQUIVALENCE, taken strictly, says fund as short as possible and pay no term premium. In practice that is rollover risk — so governments spread maturities and issue on a predictable calendar.

OECD (38 members), end-2021: weighted average maturity 7.6y against 6.3y in 2007; long-term ≈ 85% of debt outstanding; annual issuance split evenly short vs long.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Sovereign Debt", Exhibit 1 and OECD maturity data, pp. 107–112.*

# Developed-market sovereigns issue "default-risk-free" debt with unconstrained maturity access; emerging-market sovereigns face currency restrictions and external-debt stress

*A DM investor buying EM external debt has no direct FX exposure — but plenty of indirect exposure, because repayment depends on the issuer earning foreign currency.*

| Dimension | Developed-market sovereign | Emerging-market sovereign |
|---|---|---|
| Economic base | Strong, stable, well-diversified | Higher growth, less diversified |
| Currency of issuance | Reserve currency (USD, EUR, JPY, GBP…) | Restricted ccy; external FX debt |
| Maturity access | Unconstrained across the curve | Capped by domestic-ccy depth |
| Tax base | Broad and recurrent | Often commodity-concentrated |
| Default risk | Near zero in domestic currency | Material on external debt |
| Fiscal transparency | High — stable, published, rule-based | Less stable, more political |

**SRI LANKA — FIRST ASIA-PACIFIC EXTERNAL DEFAULT IN MANY YEARS, MAY 2022**

Debt rose from 87% to over 100% of GDP by 2020 as infrastructure was financed by global markets, supranationals and trading partners. Tourism and remittances collapsed that year; the 2022 inflation and commodity shock finished it. A missed USD 78m foreign-currency interest payment, after a 30-day grace period, was the default. 2021 external debt: market bonds 47%, ADB 13%, Japan 10%, China 10%, World Bank 9%, other 9%, India 2%.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, DM vs EM sovereign issuers and the Government of Sri Lanka external debt box, pp. 108–110.*

# Sovereign debt is sold at public auction — a single-price auction gives every winner the cut-off yield, a multiple-price auction pays each winner their own bid

*Single-price cuts yield volatility and the fear of the winner's curse — which is why the RBI moved most of its tenors to single-price in July 2021 after a run of partially failed auctions.*

| Dimension | Single-price (uniform-price) | Multiple-price (discriminatory) |
|---|---|---|
| Price paid by winners | All winners pay the cut-off price | Each winner pays their own bid |
| Ranking of competitive bids | Ranked from the lowest yield up | Same ranking, different pricing |
| Cut-off ("stop") yield | Highest yield that fills the offer | Same cut-off defines who wins |
| Non-competitive bids | Accepted in full at the auction price | Accepted in full; auction price |
| Yield volatility | Lower — one rate for all winners | Higher spread of winning prices |
| Distribution and cost | Broader; may lower the cost of funds | Narrower — large bids win at bid |

**CFA LM 5 EXHIBIT 2 — MAS 30-YEAR SGS AUCTION, 21 SEP 2021**

SGD 2.6bn offered, SGD 4.1bn bid (1.58× cover). Cut-off 1.95% / 98.303; median 1.89%; average 1.84%.

**Coupon = cut-off rounded down to nearest 0.125% = 1.875%; all winners pay 98.303**

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Sovereign Debt Issuance and Trading", Exhibit 2 (Singapore) and the Reserve Bank of India box, pp. 115–116.*

# Primary dealers must bid competitively at every auction, and much of the demand behind them is "non-economic" — which is why sovereign yields sit below credit curves

*This bid is structural and persistent. It is a large part of why the "default-risk-free" yield is genuinely low, and why on-the-run issues carry a liquidity premium.*

| Buyer | Why they hold it | Nature of the bid |
|---|---|---|
| Primary dealers | Must bid in every auction; the CB's OMO counterparty | Obligation |
| Domestic central bank | Conducts monetary policy — outright buys, repo, QE | Policy, not return |
| Foreign central banks | Hold FX reserves in reserve-currency sovereigns | Reserve holding |
| State and local governments | Face restrictions on holding other securities | Rule-constrained |
| Banks | Must hold, or are incentivised to hold, for regulation | Regulation |
| Insurance companies | Same regulatory pull, plus liability matching | Regulation + ALM |

## THE DOLLAR'S SHARE OF THE SYSTEM

The US dollar is about 60% of global FX reserves and roughly half of international trade, international loans and global debt securities, and is on one side of nearly 90% of FX transactions. The euro, yen, sterling, Swiss franc, Australian dollar and renminbi are also reserve currencies. This structural demand pulls sovereign yields below where credit and rate fundamentals alone would put them.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, primary dealers, secondary sovereign trading and reserve currencies, p. 117.*

# Non-sovereign issuers fall into three families — agencies backed by a sovereign, local authorities backed by taxes or project revenues, and supranationals backed by member states

*All three sit between pure sovereign and pure corporate credit — and their credit quality is driven entirely by how predictable and stable the repayment source is.*

| Issuer type | Primary repayment source | Support behind it | Example |
|---|---|---|---|
| Government agency | Agency operations / guaranty fees | Sovereign backing | AAHK, Ginnie Mae |
| Local GO bond | Local tax revenues | Tax base only | Ontario |
| Local revenue bond | User fees / tolls from the project | Project only | Toll authority |
| Quasi-gov / supra JV | Project cash flows | Sponsor consortium | PT IIF |
| Supranational | Member capital and loan repayments | Member-state support | World Bank, ADB |

## RANKING, AND TWO ISSUES WORTH KNOWING

Supranationals carry the HIGHEST credit quality of the three: member states share control and provide implicit and explicit support. Agencies borrow at a yield close to their sovereign guarantor, but neither AAHK nor Ginnie Mae gets the full sovereign liquidity premium.

Ontario 1.55% 2029, CAD 2.75bn Global Green Bond — a GO bond repaid from provincial taxes, with proceeds tracked to five eligible green sectors. ADB 7.80% 2034 Global MTN — principal fixed in IDR but paid in USD at a rate set two days before each payment date, so the investor keeps the currency risk.

*Source: CFA Institute, Curriculum Vol 6, Learning Module 5, "Non-Sovereign, Quasi-Government, and Supranational Agency Debt", pp. 118–121.*

KEY TAKEAWAYS

# Deck 13 — Fixed Income I — Instruments, Markets & Issuance

- Six features specify any bond: issuer, maturity, principal, coupon, seniority, contingency provisions. Annual coupon = par × rate.
- FRN coupon = MRR + a spread fixed at issuance. Antelas EUR 250m at MRR+250bps, MRR 0.25% → EUR 1,718,750 a quarter.
- Bullet, amortizing and balloon differ only in when principal returns. $A = rP / [1 - (1+r)^{-N}]$; the interest share falls over time.
- Cross-default is affirmative; leverage and dividend tests are negative. BRWA (IG) has three negatives; VIVU (HY) adds an incurrence test.
- A call belongs to the ISSUER and caps the price; a put belongs to the INVESTOR and floors it. Callables yield more, putables less.
- Repo: the seller keeps ownership and the coupons, the lender earns the rate. 102% margin on 100m UST → 98.04m loan, 1.96% haircut.
- Single-price auctions give every winner the cut-off yield: Singapore 1.95% → coupon 1.875%. Multiple-price pays each winner their bid.
- Non-economic demand — central banks, FX reserves, regulated holders — pulls sovereign yields below pure credit fundamentals.